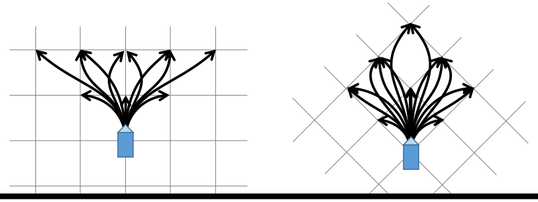
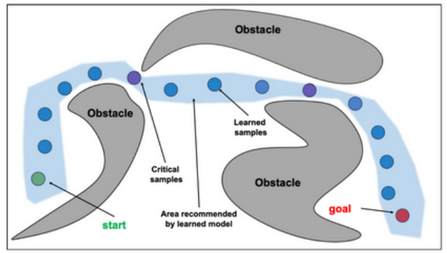
Planning as Search

- Discretize a continuous state space through random sampling → find shortest path over search tree → apply time-parameterization [1] [13]
- Sample random configurations → connect with steering function [8]
- Sample random state and action → apply forward propagation [9]

[12, 15, 16]



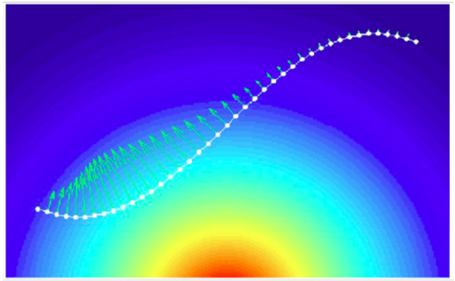
- Generate task-biased state space samples
- ML model to verify trajectory feasibility [13]
- Learning-based dynamics for steering [14]



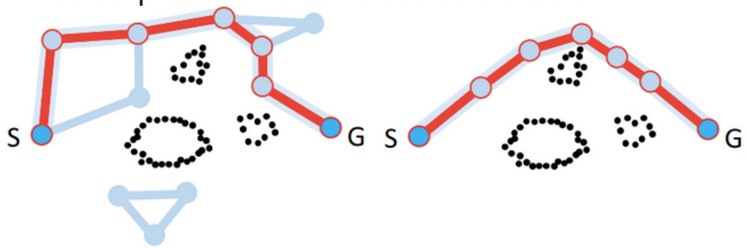
[3]

Planning as Optimization

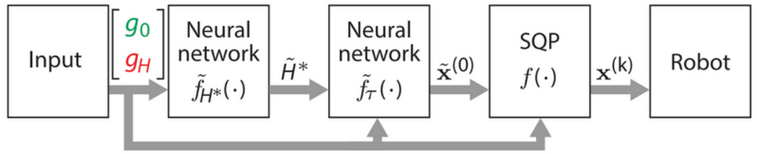
- Setup equality/inequality constraints → start with linear interpolation seed → warp for constraint satisfaction [2]



- Hybrid methods: start with geometric seed path → fit controls to solution [4]

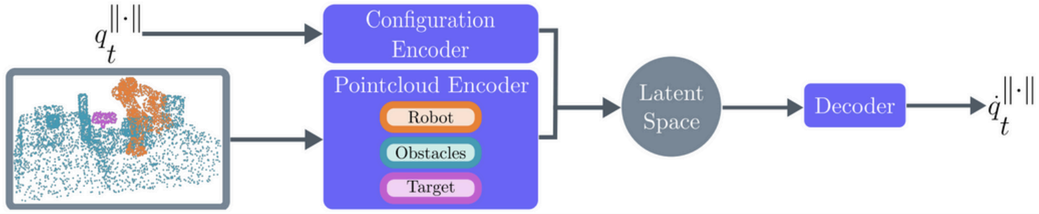


- Use a neural network to generate an informed seed for trajectory optimization [5] [10]
- Model a dynamics constraint using machine learning [3]



Planning as Inference

- Next-token (autoregressive) approaches: train a model to predict the next likely configuration [6]
- Full-sequence models: train a neural network to model distribution of trajectory space → infer solution for any new start/goal pair and environment [7] [11]



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[10] **Pasricha, Anuj** et al. "Diffusion-Based Task-Agnostic Seed Generation for Trajectory Optimization with Dynamic Constraints." Proposed work for IEEE RA-L 2025.
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